

CHAPTER 21 THE ROLE OF PRESSURE-DEPENDENT VISCOPLASTICITY AND VOLUMETRIC DILATATION IN ENERGETIC MATERIALS AT INTERMEDIATE STRAIN RATES

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1 ABSTRACT

The mechanical behavior of energetic aggregate materials (polymer bonded explosives, propellants, etc.) at low to intermediate strain rates is a complex function of the material constituents, manufacturing methods, and microstructure morphology. Shear-induced volumetric dilatation, similar to the characteristic behavior of granular materials and soils, can play a key role to open up porosity in the microstructure, particularly when the material is under some degree of confinement. We present a continuum modelling approach that accounts for this phenomena via a pressure-dependent viscoplastic extension to the classical ViscoSCRAM [1] constitutive model and use it to study the role of shear-induced dilatation in confined and unconfined material at a variety of strain rates. Modeling results are compared with intermediate strain rate Split Hopkinson Pressure Bar experiments and show good agreement between predicted and measured stresses.

2 INTRODUCTION

The mechanical behaviour of energetic aggregate materials (polymer bonded explosives, propellants, etc.) at low to intermediate strain rates is a complex function of the material constituents, manufacturing methods, and microstructure morphology. Shear-induced volumetric dilatation, similar to the characteristic behaviour of granular materials and soils, can play a key role to open up porosity in the microstructure, particularly when the material is under

some degree of confinement. We present a continuum modelling approach that accounts for these phenomena via a pressure-dependent viscoplastic extension to the classical ViscoSCRAM constitutive model and use it to study the role of shear-induced dilatation at a variety of strain rates in an HMX-based explosive.¹ Modelling results are compared with intermediate strain rate Split Hopkinson Pressure Bar experiments and show good agreement between predicted and measured stresses.

2.1 Hypothesized damage mechanisms

The mechanical behaviour of energetic aggregates is complex and has been widely studied. Macroscale behaviours observed include strain rate and temperature dependence, viscoelastic behaviour in the polymer binder, inelastic damage evolution and yielding, pressure-dependence, and tension-compression asymmetry.² These features of the mechanical response are driven by material properties of the energetic and binder constituents, their proportions in the composite material, presence of pre-existing defects, and the various damage mechanisms that evolve at the mesoscale as load is applied.

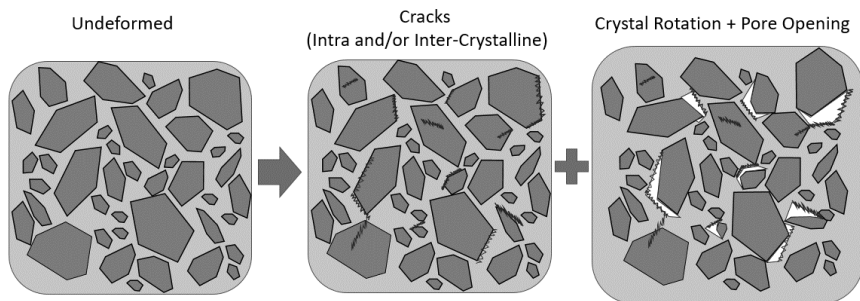


Figure 1 *Schematic of hypothesized damage mechanisms. The crystal rotation and porosity opening are associated with a positive volume strain.*

In the current work, we hypothesize that mechanical damage evolves as a combination of cracking and porosity opening mechanisms (see Figure 1). For simplicity, no distinction is made here between intra-granular and inter-granular cracking at the interface between the energetic crystals and polymer binder (debonding). Once these cracks have formed, the partially debonded crystals are free to rotate within the microstructure as loading continues. This mechanism causes additional porosity to open up internally and is associated with positive macroscale volume strain. Each of these mechanisms has been observed experimentally, but we note that all may not be active simultaneously, and one may dominate over another in different loading regimes.^{3, 4} These are also not the only damage mechanisms that may be present, but even this idealized view of the underlying mesoscale processes provides valuable insight when choosing macroscale model forms with the goal of predictive, physics-based modelling.

3 VISCOPLASTIC-VISCOGRAM MATERIAL MODEL

Based on the mechanisms discussed in Section 1, we present an extension to the classical ViscoGRAM¹ constitutive model to include pressure-dependent viscoplasticity and volumetric dilatation. The Viscoplastic-ViscoGRAM model draws on elements of the papers by Buechler as well as standard viscoplasticity theory from Kojic and Bathe.⁵⁻⁷ The linear viscoelasticity and damage mechanics of the ViscoGRAM model remain largely unaltered but are embedded within a yield surface plasticity model. Details of the model theory and its implementation within the SIERRA-Solid Mechanics finite element code suite are given by Vignes and Long with a brief summary here.⁸

The deviatoric strain ε_{ij} is additively decomposed into a viscoelastic component ε_{ij}^{ve} , a damage component ε_{ij}^D , and a viscoplastic component ε_{ij}^p as

$$\varepsilon_{ij} = (\varepsilon_{ij}^{ve} + \varepsilon_{ij}^D) + \varepsilon_{ij}^p. \quad (1)$$

The viscoelastic and damage components are computed as in classical ViscoGRAM¹ and account for linear viscoelastic behavior of the binder and a statistical crack mechanics (GRAM) framework that represents the nucleation and growth of microcracks (damage). The parameter, c , represents an averaged microcrack size that evolves based on brittle fracture mechanics including stable and unstable crack growth regions. From the GRAM portion of the model, we define an alternate form of damage parameter, D , that depends on the crack size, c , and crack normalization, a , parameters as

$$D = \frac{\left(\frac{c}{a}\right)^3}{1 + \left(\frac{c}{a}\right)^3}. \quad (2)$$

This follows the form of a traditional continuum mechanics damage parameter that varies between 0 for the undamaged state ($c = 0$) and asymptotes to 1 for fully damaged material.

The viscoplastic model is based on an overstress assumption such that a stress point may lie outside the yield surface $f(\sigma_{ij}, \varepsilon_{ij}^p, D)$ and move towards the yield surface with time. Viscoplastic flow occurs until the stress point returns to the yield surface, and in general, flow can be specified as either associative or non-associative. For the current work, we chose Drucker-Prager forms for the yield surface and flow surface with potentially different fitting constants defined, respectively, as

$$f(\sigma_{ij}) = \sigma_e + A \cdot \sigma_m - \sigma_y \text{ and } g(\sigma_{ij}) = \sigma_e + B \cdot \sigma_m - \sigma_y, \quad (3)$$

where σ_e is the equivalent stress given by

$$\sigma_e = \sqrt{\frac{3}{2} s_{ij} s_{ij}}, \quad (4)$$

and where σ_m is the mean stress, σ_y is the yield stress, and A and B are fitting constants. The flow rule is selected to be

$$\dot{\lambda} = \frac{1}{\tilde{\tau}} \left\langle \frac{f(\sigma_{ij})}{\sigma_0} \right\rangle^{\tilde{m}}, \quad (5)$$

where $\tilde{\tau}$ is the viscoplastic relaxation time, σ_0 is a normalizing constant, and $\tilde{m}$ is a fitting exponent.

The Viscoplastic-ViscoSCRAM model was calibrated in two stages. An initial calibration for the ViscoSCRAM parameters from Buechler and Luscher was used.⁹ The viscoplasticity parameters were then calibrated independently. For simplicity, we assumed associative flow such that $g(\sigma_{ij}) = f(\sigma_{ij})$, which results in $A = B$. We also assume rate independence for plastic flow by setting $\tilde{m} \gg 0$ and $\tilde{\tau} \ll 1$. This leaves only two independent parameters, σ_y and A , which were calibrated with triaxial data for an HMX-based explosive.¹⁰

4 RESULTS AND DISCUSSION

4.1 Uniaxial compression study

A simple finite element model of a cylinder in uniaxial compression with displacement control boundary conditions was used to study the model behaviour at different strain rates. These simulations were done with an implicit, quasi-static solver, and no dynamic effects (mass inertia, stress wave propagation, etc.) were considered.

Figure 2 shows the predicted stress strain curve, damage parameter (D) evolution, equivalent plastic strain, and volume strains as a function of engineering axial strain for strain rates ranging from $1 \times 10^{-4} \text{ s}^{-1}$ – 100 s^{-1} .

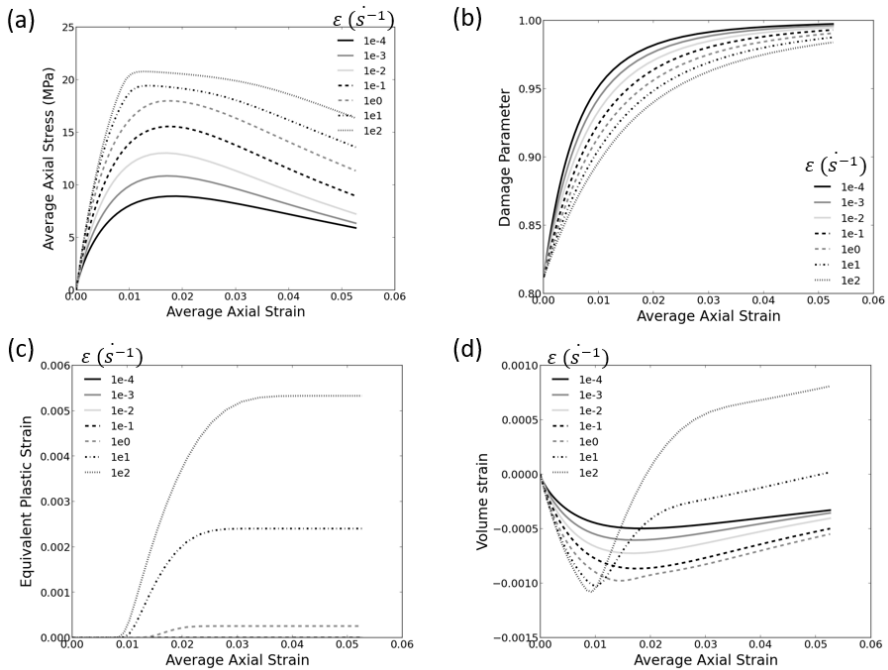


Figure 2 Model predictions for uniaxial compression of an HMX-based explosive at several different strain rates. (a) engineering stress-strain curve, (b) damage parameter, (c) equivalent plastic strain, (d) volumetric strain.

A strong rate dependence is predicted such that both the initial stiffness and the peak stress increase with increasing strain rate. This is consistent with observed behaviour of various HMX-based explosives.² The rate dependence in model predictions results because both the linear viscoelasticity and the crack growth formulations in the model are independent functions of strain rate. From Figure 2b, the damage parameter approaches 1 (fully damaged state) more slowly at the higher strain rates. Plasticity, demonstrated by the presence of non-zero equivalent plastic strain, is only seen at the higher strain rates of 1 s^{-1} and above and correlates with the sharp increase in volumetric strain that occurs around 1% axial engineering strain in those cases. Since we have purposely chosen the plasticity mechanism to be rate independent, we can infer that some combination of the viscoelasticity and damage mechanisms (both of which are rate dependent) results in activation of the plastic flow at higher strain rates only.

This simple case study demonstrates the complex interaction between the various viscoelastic, crack-driven damage, and plasticity mechanisms of the model. At the slower strain rates below 1 s^{-1} , cracking damage begins to accumulate slightly before the peak stress. This reduces the load carrying capacity of the material and results in strain-softening and peak stresses well below the plastic yield strength. At the faster strain rates $\geq 1 \text{ s}^{-1}$, different portions of the viscoelastic relaxation spectrum are activated which stiffen the

initial elastic response and allow the material to reach stresses above its yield point such that plasticity occurs in addition to cracking damage. The plastic deformation acts as a second inelastic mechanism that reduces material strength and produces positive volume strain such as could occur from particulate debonding and rotation.

4.2 Split Hopkinson pressure bar dynamic loading

Intermediate strain-rate impact behaviour of the HMX-based material was studied with a Split Hopkinson Pressure Bar (SHPB). A momentum trap was used for passage of a single compressive wave prior to sample recovery. Details of these experiments are given by Brown and Cooper et al., and results from a representative test with a strain rate of 365 s^{-1} are shown here for comparison with the model.¹¹

A simplified finite element model of the SHPB test with partial length incident and transmitted bars is shown in Figure 3. A velocity boundary condition of 4.6 m/s was applied to the end of the partial-length incident bar to approximate the strain rate measured during the experiment. The end of the transmitted bar was given fixed displacement boundary conditions. Frictionless contact was used at the interfaces between the bars and the explosive sample. The modeled bar lengths were carefully chosen so that the stress wave traveling along the bars only interacted once with the sample. Further details of the finite element model set-up are given in Brown and Cooper et al.¹¹

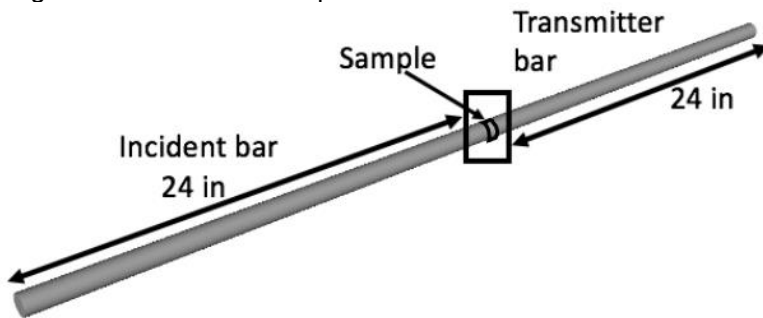


Figure 3 *Split Hopkinson pressure bar model. Further model and test details are available in Brown and Cooper et al. Reproduced from ref. 11 with permission from JANNAF.*

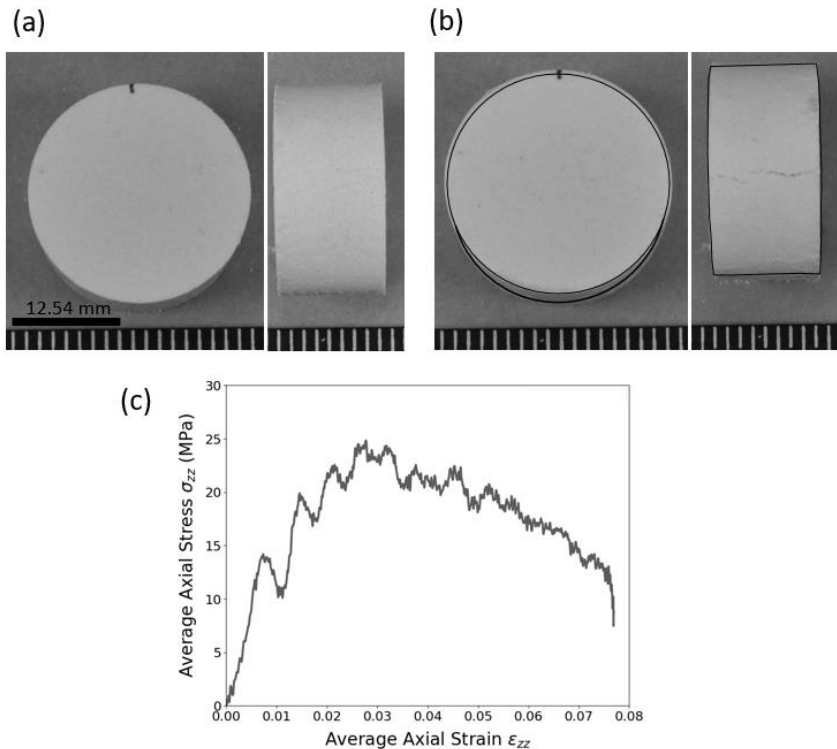


Figure 4 *Experimental Results: (a) pre-test images of the top and right sides of the SHPB sample, (b) post-test images of the same views, (c) engineering stress-strain curve calculated from the test data. All images have the same scale.*

Pre-test (a) and post-test (b) pictures of the explosive sample are shown in Figure 4 with the engineering stress-strain curve (c) calculated from the test strain gauge signals using standard one wave analysis. We note that the primary goal of this test was to induce damage into the bulk material from intermediate strain rate impact, and due to its size for this purpose, the sample was not strictly in stress-equilibrium for the duration of the test. Thus, the stress-strain curve shown in Figure 4 gives a general idea of the material response under these non-equilibrium conditions rather than rigorous, material property data. Image analysis (Image J) was used to measure sample dimensions from the pre- and post-test images and to superimpose the outline of the pre-test sample onto the post-test images (black outlines shown in Figure 4b).¹² Assuming that the sample is a right circular cylinder both pre- and post-test, we computed the approximate volume strain of the sample $\Delta V/V_0$ to be 4.6%.

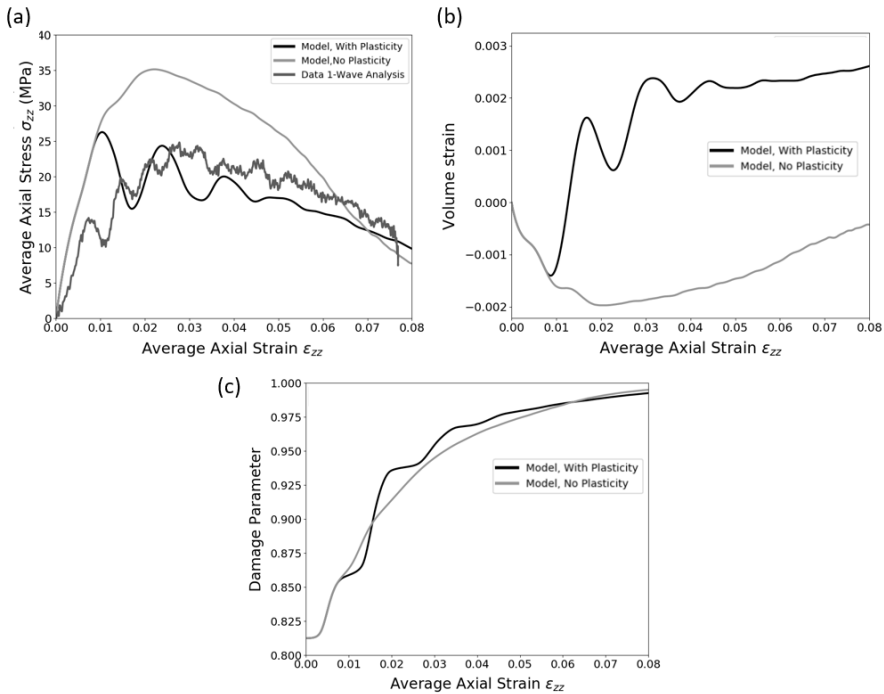


Figure 5 SHPB model predictions with and without viscoplasticity: (a) stress-strain curve compared with experimental result (b) volumetric strain, and (c) damage parameter, D .

Model predictions of the stress-strain curve, volumetric strain, and damage parameter with and without viscoplasticity are shown in Figure 5. The predictions without plasticity were achieved by setting a very high yield stress such that plasticity never occurs. The experimental stress-strain curve from Figure 4c is superimposed on the modelling result for comparison. The initial slope of the stress-strain curve predicted by both versions of the model is noticeably stiffer than the data. This is not surprising, however, due to the simplified nature of the SHPB model and its use of velocity boundary conditions to approximate the stress wave traveling along the incident bar. Experimentally, a force boundary condition occurs at the interface between the striker and incident bar, but this was not modelled explicitly to reduce model size and computational cost.

The peak stress from the full Viscoplastic-viscoSCRAM model shows good quantitative agreement with the data, and the general shape and magnitude of the stress-strain curve is also a good match. This contrasts with the model without plasticity, which overpredicts the peak stress by ~ 10 MPa and has a much steeper negative slope in the post-peak strain-softening region compared with the data. Additional comparison between the two models shows that the model without plasticity predicts a net negative volume strain (shrinkage) whereas the model with plasticity predicts a sharp jump in positive volume strain

when plastic yield is reached and shows net positive volume strain (dilatation). The final values of the damage parameter are similar for both models but show different evolution behaviour to reach this state.

These results indicate that the viscoelasticity and cracking damage mechanisms alone are not enough to capture all the inelastic deformation mechanisms that drive mechanical behaviour at intermediate strain rate conditions. The addition of a simple pressure-dependent viscoplasticity mechanism to the model improves predictions of the peak stress and strain-softening behaviour compared to the experiment. It also allows for shear-induced dilatation that results in net positive volume strain to be captured. The experimental sample also had a net positive volume strain when measured post-test as well as visible cracking on the surface (see Figure 4). We attribute the measured positive volume strain to the combination of cracking and porosity opening mechanisms discussed in Figure 1. Additionally, evidence of such behaviour has been reported previously in SHPB tests by Ravindran et. al.⁴ Even with the improvement seen when plasticity is added to the model, the predictions of volume strain are still an order of magnitude less than the experimentally estimated value. We note that exploring model forms with non-associative flow would allow more dilatation to be predicted as in granular media and geological materials.

Figure 6 shows spatially averaged characteristics of the plasticity response predicted by the full Viscoplastic-ViscoSCRAM model, and Figure 7 shows timeseries contour plots of axial stress, damage parameter, and equivalent plastic strain evaluated at the four marked points on the stress-strain curve in Figure 6. The two peaks in the stress-strain curve at points A and C correspond to non-zero plastic work rates and sharp increases in equivalent plastic strain. The contour plots in Figure 7 show that these correspond to stress waves propagating in the radial directions and interacting with the circumferential cylinder surface.

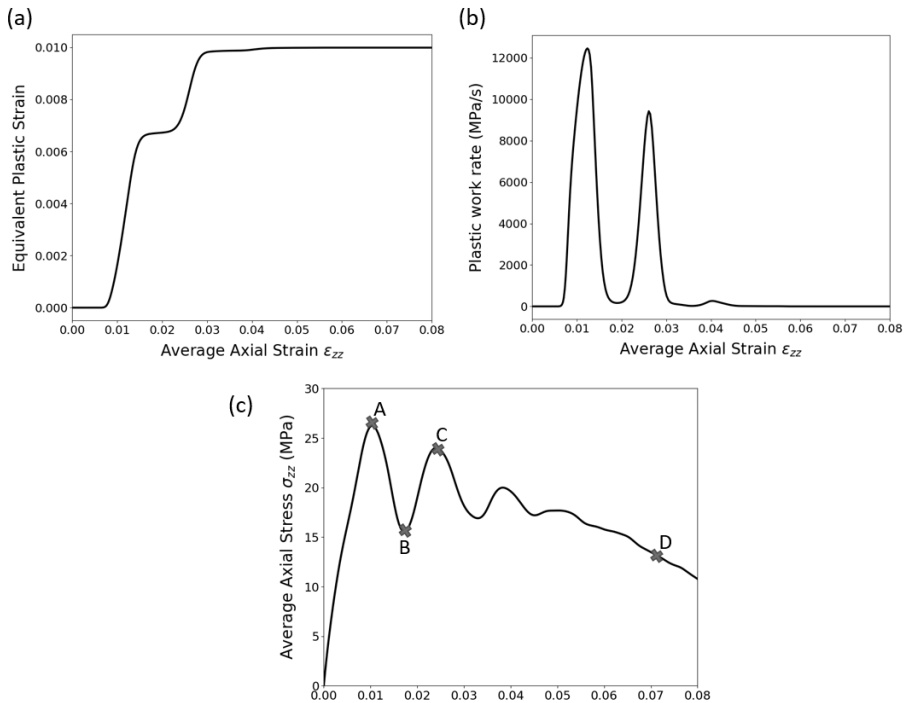


Figure 6 Viscoplastic-ViscoSCRAM model predictions of spatially averaged values for (a) equivalent plastic strain (b) plastic work rate, and (c) stress-strain curve with markers showing oscillations due to radial stress waves (A-C) and late damage evolution (D).

The interplay between the cracking damage mechanism and plasticity mechanism under these spatially and time-varying stress states is seen from Figure 7. At the initial peak stress (point A in Figure 6c, Figure 7a), a ring of material at the outer surface is damaged via the SCRAM cracking mechanism. Since the stresses in this ring of damaged material are reduced compared to the rest of the sample, the initial plastic strain occurs just inside the damaged region. At point B, the radial stress wave has reflected from the outer surface and has reduced the axial stresses in the sample interior (but presumably created tensile radial stresses). This complex stress state corresponds to additional cracking damage in the sample center and additional plastic strain in the areas where cracking damage has not further reduced the stress. This process is repeated through the additional oscillations (point C) and results in the final state shown at point D where the stresses are lower throughout the sample and the material has pervasive cracking damage and plastic strain.

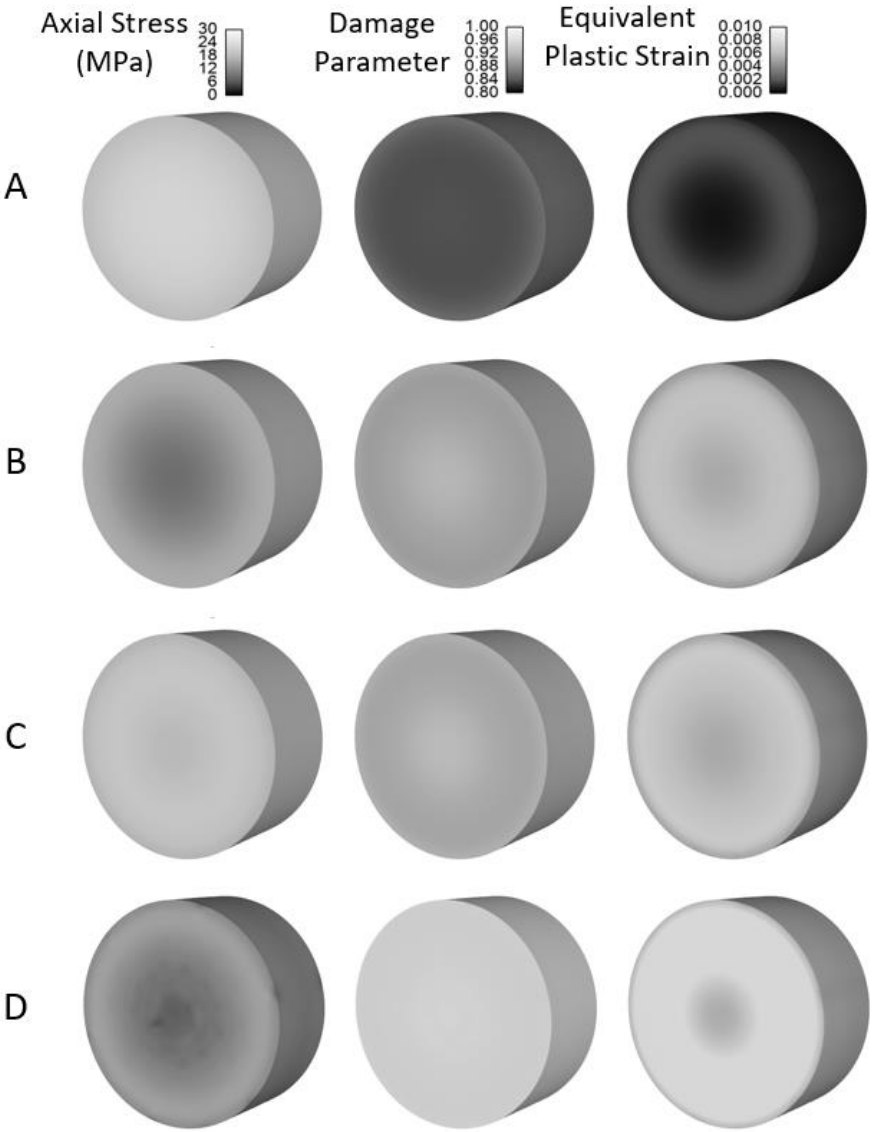


Figure 7 Timeseries of contours for (left) average axial stress, σ_{zz} (middle) damage parameter D , (right) equivalent plastic strain.

The damage parameters close to 1.0 at point D indicate that the material is heavily damaged with the statistical representation of microcracks given by the SCRAM model. This does not necessarily indicate formation of macroscale, connected crack networks, however. Interpretation of the cracking damage

parameter and plastic yielding to determine a material failure criterion is beyond the scope of this study, but a rich area to explore in future work.

The final material state predicted at point D seems qualitatively similar to the observed experimental sample, which showed visible cracks around its circumferential surface that ran in the axial loading direction. Further work is ongoing to better characterize the material state throughout its interior after such intermediate strain rate impacts and determine if the inelastic mechanisms modelled here (SCRAM cracking damage and pressure-dependent plasticity) are truly representative of the material's mesoscale damage processes.

5 SUMMARY AND CONCLUSIONS

A Viscoplastic-ViscoSCRAM constitutive model that includes a pressure-dependent, Drucker-Prager yield function was used to predict uniaxial compression behaviour at a variety of strain rates in an HMX-based explosive. Modelling results with and without the viscoplasticity mechanism were compared with intermediate strain rate Split Hopkinson Pressure Bar experiments. Model predictions that included viscoplasticity showed good quantitative agreement with experimentally measured peak stress and qualitative agreement with measured positive volume strains (dilatation). Without the viscoplasticity mechanism, the model over-predicted peak stress compared to the experiment and predicted net negative volume strain.

These results indicate that a mechanism to allow inelastic, shear-induced dilatation is needed in continuum constitutive models for the HMX-based plastic bonded explosive studied here, particularly for dynamic strain rates. We hypothesize that the underlying mechanism is rotation of debonded crystals in the microstructure that opens up additional porosity and leads to shear-induced dilatation akin to that seen in soils and other granular media. Further work is needed to fully understand the stress states that induce this behaviour, how best to represent its effects in continuum models, and applicability to the broader class of plastic bonded explosives with different binder materials and volume fractions, etc.

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